UNIVERSITY OF CHINESE ACADEMY OF SCIENCES

DOCTORAL THESIS

The study of Higgs properties in the four-lepton final state at CMS

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Chapter 1

Probing four lepton final-state for Higgs Cross section measurement

1.1 Introduction

The ATLAS and CMS collaborations of CERN announced the discovery of much awaited Higgs boson in early July 2012 [1, 2] verifying the underlying concept of electroweak symmetry breaking (EWSB) in the SM of particle physics. This triggered essential precise studies of this particle to confirm its nature and to discover some new physics in case of some deviation while studying its properties. Later on, these experiments have also studied its comprehensive properties [3, 4, 5, 6] which are found to be consistent with the SM Higgs. There is a legacy of measurements on this topic from CMS and ATLAS collaborations in different LHC data taking eras for different decay cahannels of Higgs which can been studied in Ref. [**Khachatryan:2015yvw**, **CMS:2015hja**, **CMS:2016rqf**, **Sirunyan:2017exp**, **CMS:2018hhg**, **atlas-conf-2019-029**, **atlas-conf-2019-025** , **atlas-conf-2019-032**, **CMS:2019drq**, 7, 8, 9, 10, 11, 12, 13]. However, these measurements can be improved with newer data at higher luminosities and improved statistical techniques.

The cross section measurements (integrated and differential) are performed using collisions data of CMS (LHC) corresponding to 137.1 fb^{-1} integrated luminosity at $\sqrt{s} = 13$ TeV. The studied differential observables are sensitive to production mechanism of Higgs. Further details follow in next sections.To minimize the model dependence, results are extracted in the fiducial volume that closely matches with the detector geometry as shown in Figure 1.1 [**ref_fiducial**]. Conribution of other model dependence is extracted by repeating the measurements for SM and other exotic models. The differences are reported as systematic uncertainty. All measurements includes the corrections for the finite detection efficiency and resolution effects, and are extracted assuming $m_H = 125.09$ GeV, as measured from $H \to 2\gamma$ and $H \to 4\ell$ decay modes at CMS experiment of LHC [14].

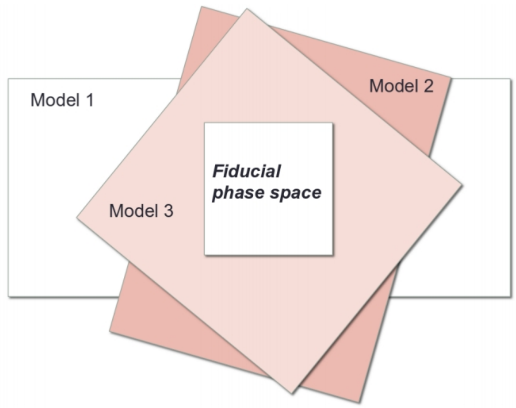

FIGURE 1.1: Fiducial phase definition among different models.

1.2 Datasets and Triggers

The analysis uses full Run 2 (2016, 2017 and 2018) CMS data corresponding to $137.1\ \mathrm{fb}^{-1}$ intergrated luminosity.

Used datasets are listed and documented in **[CMS:2019chr]** with the corresponding integrated luminosities. The analysis relies on four different primary datasets (PDs), *DoubleMuon*, *MuEG*, *EGamma*, and *SingleMuon*, each of which combines a certain collections of HLT paths. Events are collected to avoid duplications from different PDs, using:

- EGamma if they pass the diEle or triEle or singleElectron triggers,
- DoubleMuon if they pass the diMuon or triMuon triggers and fail the diEle and triEle triggers,
- MuEG if they pass the MuEle or MuDiEle or DiMuEle triggers and fail the diEle, triEle, singleElectron, diMuon and triMuon triggers,
- SingleMuon if they pass the singleMuon trigger and fail all the above triggers.

The HLT paths used for full Run 2 collision data is listed in **[CMS:2019chr]** and also in the Appendix.

1.2.1 Trigger Efficiency

Efficiency in data of combination of triggers used for the anlysis is measured by considering 4ℓ events triggered by single lepton triggers. Among the reconstructed four leptons, is geometrically associated with an object that passes at least one single electron or muon triggers. Rest of three leptons are regarded as “probes". In a 4ℓ event, the possible tag-probe pairs are 4 and are counted in denominator while computing efficiency. For each of three leptons which are probes, all matching trigger filter objects are collected. Then the matched trigger filter objects of the three probe leptons are combined in attempt to reconstruct any of the triggers used in the anlysis. If any of the analysis triggers can be formed using the probe leptons, probes are call passing probes which are countedas numerator for the efficiency measurement.

This technique is used to measure efficiency on MC and the comparison with data is exploited to estimate the reliability of simulation. Fig. 1.2 shows the measured efficiency both in data and MC (2018) varied with minimum p_{T} of leptons. MC efficiency seems to describe data well within the uncertainties.

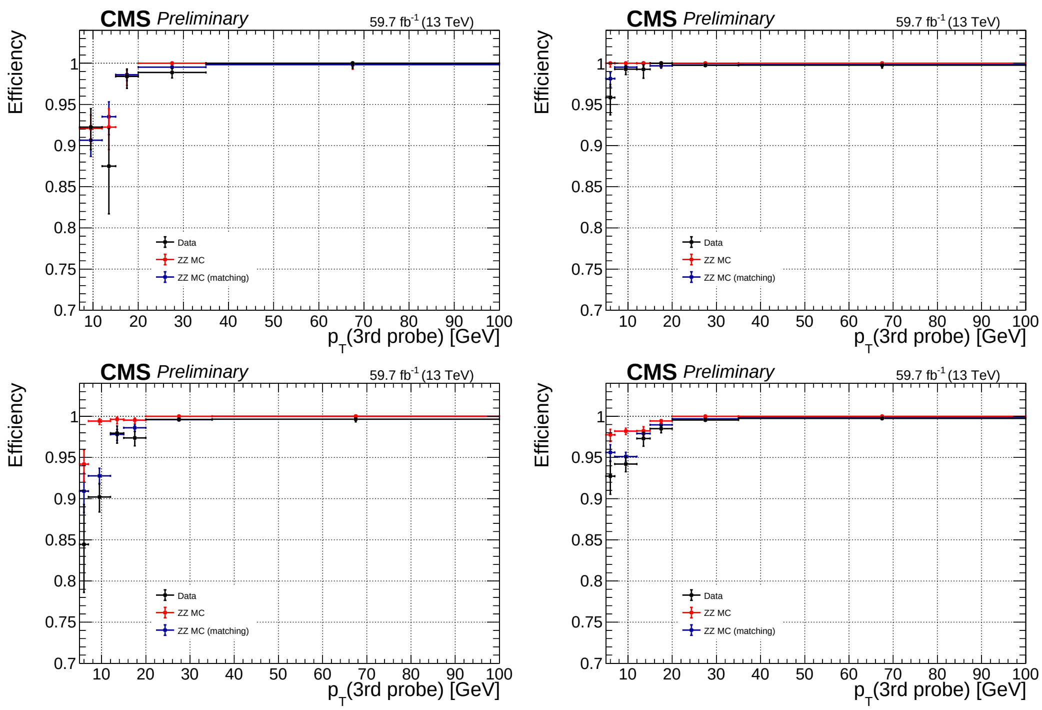

FIGURE 1.2: Measured trigger efficiency in 2018 data where the event collection is made using single electron and muon triggers: $4e$ (top left), 4μ (top right), $2e2\mu$ (bottom left) and 4ℓ (bottom right) final states.

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